

Self-Evolving Deep Research via Joint Generation and Evaluation

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Abstract

Large Language Models (LLMs) have become increasingly adopted in daily applications, with deep research standing out as a particularly important capability. Unlike traditional question-answering (QA) tasks, deep research report generation lacks definitive ground-truth, making reward design inherently unverifiable and limiting effective reinforcement learning. Existing approaches mitigate this challenge with LLM-as-a-judge and query-dependent evaluation rubrics, but they still rely on static evaluators that cannot adapt their standards as the solver improves, leading to insufficient and eventually saturated optimization pressure. We address this limitation with a self-evolving co-evolutionary training framework for deep research evaluation and generation (**SCORE**), which tightly couples an evaluator and a solver in a shared-parameter learning process. Rather than treating generation and evaluation as isolated modules, we leverage their intrinsic connection to enable joint improvement within a single shared-parameter model. To restrict this process, we introduce a meta-harness, which dynamically controls the evaluation environment based on solver performance, encouraging valid evaluation dimensions and sufficiently deep evaluator search. Extensive experiments on deep research benchmarks demonstrate consistent improvement in report generation quality, showing that co-evolving evaluation and generation is a promising direction for training open-ended research agents.

1 Introduction

Owing to their extraordinary capabilities, Large Language Models (LLMs) have become deeply integrated into various complex domains, spanning autonomous driving [48], software development [12], multimedia generation [33, 46, 54] and so on [49, 26]. As the demand for comprehensive information retrieval and synthesis grows, enhancing the deep research capabilities of LLMs has attracted increasing attention from researchers. Several search agents, such as GPT-Researcher [1] and AgentCPM-Explore [5], have recently demonstrated remarkable proficiency in these multi-hop research tasks. Furthermore, previous works like Search-o1 [27] and SSP [31] have shown that applying Reinforcement Learning (RL) to these models can substantially enhance their reasoning capabilities when integrated into agent, yielding significant performance improvements on complex Question-Answering (QA) tasks.

Differ from traditional QA tasks, training models for open-ended report generation presents two challenges. First, comprehensive reports lack a definitive ground truth, making objective reward signals inherently difficult to define [24, 15]. Since there is no absolute standard for open-ended queries, relying on human annotations for generated reports is prohibitively expensive, making objective and scalable reward signals difficult to construct. Relying on human annotations to cover these diverse paths is prohibitively expensive and difficult to scale. Second, the evaluation criteria

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for a high-quality report are multi-dimensional and query-related [22, 13]. Single fixed evaluation metric inevitably ignore crucial dimensions including factual correctness and insight diversity for different questions. This dimensional omission results in sparse training signals and fails to provide the fine-grained credit assignment required for complex generation tasks.

To overcome these challenges in report generation, recent research proposes several novel paradigms to provide more effective optimization signals. WebThinker [28] and Step-DeepResearch [18] have applied the LLM-as-a-judge [57] to evaluate report quality during the training process. AdaRubric [11] proposes a framework that dynamically generates task-specific evaluation criteria based on the input instructions. Mix-GRM [52] utilizes Reinforcement Learning with Verifiable Rewards (RLVR) to automatically adapt between breadth-first (B-CoT) and depth-first (D-CoT) evaluation strategies according to distinct task demands. However, a fixed external evaluator lacks the ability to dynamically adjust its evaluation standards, and therefore fails to provide sustained optimization pressure as the solver improves.

As empirically observed in Figure 1, LLMs’ generation quality is positively correlated with its evaluation proficiency. Building upon this insight, we hypothesize that generation and evaluation abilities can be mutually reinforcing and should be optimized within a unified, rather than decoupled, architecture. To this end, we propose a self-play framework in which the policy model iteratively generates and evaluate reports. By replacing a fixed external judge with a dynamic evaluator that evolves alongside the solver, our method maintains informative training pressure throughout learning. Furthermore, a Meta-Harness [23] is applied to stabilize this process by enforcing environmental constraints. Empirically, this co-evolutionary paradigm brings substantial and consistent improvements across diverse research generation tasks.

In a nutshell, our contributions are summarized as below:

- We propose SCORE, an evaluator-solver self-play framework for training deep research agents with unverifiable rewards. This framework provide dynamic, multidimensional rewards to optimize the solver’s ability in open-ended report generation, constrained by a meta-harness.
- We provide a theoretical analysis of SCORE under shared parameters. Our analysis clarifies the role of consistency in evaluator-side adaptation and characterizes the local optimization behavior of shared-parameter alternating updates.
- We show that our method improves agent performance across multiple evaluation dimensions on deep research benchmarks, and provide a systematic analysis of how parameters affect training procedure under unverifiable rewards.

2 Related Work

2.1 Deep Research

LLMs have exhibited strong deep research capabilities in recent agent studies [59, 4]. Initial frameworks, including Search-o1 [27], Search-R1 [21], and R1-Searcher [39], successfully introduced explicit reasoning process into agentic search frameworks, yielding substantial empirical improvement in multi-hop question-answering (QA) tasks. Following these advancements, researchers integrated diverse optimization algorithms into agentic training pipelines. Specifically, WebThinker [28] utilizes iterative Direct Preference Optimization (DPO) for deep research agents, whereas DeepResearcher [58] adopts Group Relative Policy Optimization (GRPO). Diverging from these algorithmic enhancements, SimpleDeepSearcher [40] argues that the quality of the training data supersedes the specific training paradigm. Beyond QA tasks, report generation tasks are inherently more complex, as they

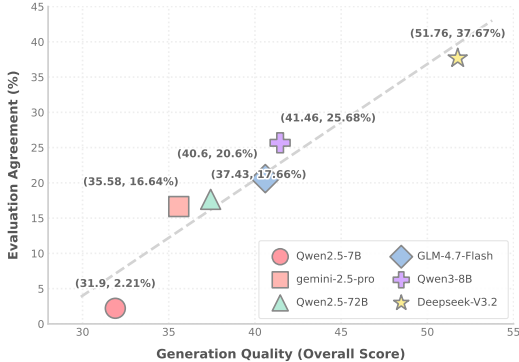


Figure 1: Empirical evidence of the positive correlation between LLMs’ generation and evaluation capabilities.

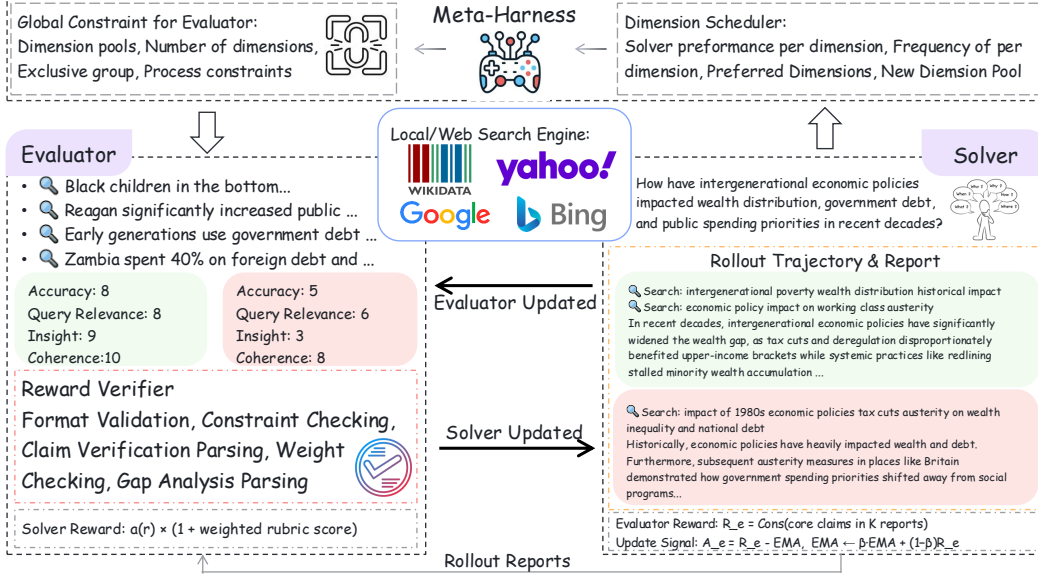


Figure 2: SCORE is controlled by Meta-Harness. The evaluator first selects evaluation dimensions from a provided dimension pool and retrieves evidence accordingly. The solver is updated using rewards derived from evaluator, while the evaluator is optimized based on the consistency of rollout reports generated by the solver. A reward verifier filters out invalid rewards before they are used for solver optimization.

involve open-ended, multi-faceted synthesis where establishing reliable ground truth for automated evaluation remains a critical bottleneck. To address this complexity, multi-agent frameworks like STORM and Co-STORM leverage preliminary knowledge and human-in-the-loop interactions during report generation [34, 20]. Later approaches such as Step-DeepResearch and AgentCPM-Report [18, 29] directly integrate model training optimized for their specific agent frameworks.

2.2 Intrinsic Rewards and Self-Play

Traditional Self-play is widely applied to agentic studies as an adversarial reinforcement learning approach [53]. AbsoluteZero, R-Zero, SeRL and SPIRAL, demonstrate the effectiveness of self-evolution in tackling complex reasoning tasks under data scarcity situation [55, 19, 30, 14]. Recent frameworks, including Search Self-play, Dr. Zero, and SPaR, demonstrate the substantial advantages of applying this technique to search agents operating within multi-hop environments [31, 50, 6]. Beyond adversarial learning paradigms like proposer-solver self-play, alternative approaches propose training solvers on filtered trajectories generated by the models themselves, as demonstrated by frameworks such as WebThinker, EvolveSearch, and EXSEARCH [28, 51, 38]. Other notable efforts include AceSearcher, which employs cooperative interaction between decomposer and solver for self-improvement [45]. Moreover, optimization strategies utilizing internal rewards provided by models themselves have been adopted by researchers in frameworks such as INTUITOR [56] and JEPO [41]. These previous works demonstrate the efficacy of internal signals in enhancing autonomous capability improvements. However, the unique characteristics of deep research task present a significant evaluation challenge. The difficulty of reliably scoring open-domain, comprehensive reports makes it highly challenging to design effective reward signals that resolve the credit assignment issue.

3 Methodology

3.1 Problem Setting

We study *deep research report generation*, where the goal is to learn a policy $\pi(r \mid q, \mathcal{E}_q)$ over candidate reports r for a given query q and environment $\mathcal{E}_q$. We assume that report quality is governed by a latent functional $Q(r; q, \mathcal{E}_q)$, and the ideal objective is to assign more probability mass to reports

with higher latent quality. Here, $q \in \mathcal{Q}$ denotes a user query, $r \in \mathcal{R}$ denotes a candidate long-form report, and $\mathcal{E}_q$ denotes the query-specific evidence and evaluation environment.

Unlike QA-style tasks, $\mathcal{Q}$ is inherently *multi-dimensional* and *query-dependent*: a report may succeed or fail along different axes, such as factual support, evidence coverage, citation grounding, contradiction handling, uncertainty calibration, and discourse coherence. As a result, long-form report generation rarely admits a single reliable verifier or a fixed scalar reward that is both sufficiently informative and broadly applicable.

We therefore formulate deep research as a query-conditioned report optimization problem with an evaluation bottleneck. In particular, the central challenge in training deep research models is how to evaluate long-form reports in a query-specific and stable manner under limited supervision.

3.2 Overview of SCORE

To address this challenge, we propose SCORE, a retrieval-grounded and externally guided evaluator-solver framework for deep research report generation as shown in Figure 2. SCORE contains four components:

1. an external evidence environment that can be queried through search and retrieval tools,
2. a fixed external *meta-harness* that shapes the query-specific evaluation environment,
3. an *evaluator* that constructs query-conditioned rubrics and produces structured report assessments,
4. a *solver* that interacts with the evidence environment, gathers supporting evidence, and generates reports under the resulting rubric.

A key design choice of SCORE is that the solver and evaluator are implemented as two functionally distinct roles over a *shared underlying actor*. This design is motivated by the substantial overlap between report generation and report evaluation, both of which require query understanding, evidence modeling, and discourse-level reasoning. We empirically examine this choice in our ablations Section 5.2.

Training proceeds through sequential alternating updates on the shared actor: the solver is optimized using evaluator-derived report rewards, while the evaluator is optimized using rubric-level auxiliary signals. Both updates are regularized by KL divergence against a reference policy to limit excessive policy drift.

3.3 External Evidence and Query-Specific Environment

For each query q , SCORE interacts with an external evidence environment through search and retrieval tools. We denote by

$$D_q^{(0)} = \mathcal{R}(q)$$

the initial evidence obtained from this environment, where $\mathcal{R}$ abstracts the retrieval interaction available to the system. In practice, this evidence environment may include web search, document retrieval, or other tool-mediated evidence sources.

Based on the initial evidence $D_q^{(0)}$, SCORE constructs a query-specific evaluation environment through a fixed external *meta-harness*:

$$\mathcal{E}_q = \text{Harness}(q, D_q^{(0)}, \mathcal{S}),$$

where $\mathcal{S}$ denotes recent training statistics or historical signals available to the harness. The resulting environment $\mathcal{E}_q$ specifies the admissible evaluation space for the current query, including candidate evaluation dimensions, structural constraints on valid rubrics, process requirements for evidence verification or comparison, and auxiliary task-specific guidance.

The meta-harness is external and non-trainable. It is implemented as a fixed external controller and is not optimized by gradient descent together with the actor. Its role is to shape the admissible evaluation space in which evaluator-solver interaction takes place.

3.4 Evaluator and Solver

Given query q and environment $\mathcal{E}_q$, the evaluator samples a query-conditioned rubric

$$(H_q, w_q) \sim \pi_e(\cdot \mid q, \mathcal{E}_q),$$

where $H_q = \{h_1, \dots, h_k\}$ denotes the selected evaluation dimensions and w_q their corresponding weights.

Conditioned on the query, the current evidence state, and the evaluator-selected rubric, the solver interacts with the external evidence environment to gather supporting information and generate a group of candidate reports

$$r_1, \dots, r_K \sim \pi_s(\cdot \mid q, D_q^s, H_q, w_q),$$

where D_q^s denotes the evidence collected by the solver and K is the number of solver rollouts sampled for the same query during training.

For assessment, the evaluator may perform constrained supplementary retrieval or verification based on the query and the generated reports, producing its own evidence context D_q^e . It then produces structured judgments over candidate reports, including dimension-wise scoring, evidence support checking, contradiction analysis, and citation verification. These structured judgments are converted into solver-side learning signals.

3.5 Shared Actor and Training Signals

SCORE implements the solver and evaluator as two roles over a shared actor with parameters θ . This design is motivated by the observation that report generation and report evaluation rely on overlapping latent capabilities, including query understanding, evidence modeling, and discourse-level reasoning. Under this view, generation and evaluation are not treated as fully separate functions, but as two role-specific mappings induced from a shared underlying representation. A shared actor therefore provides a common representational substrate for both roles, allowing evaluator-side adaptation to influence the same internal representations used by the solver. We empirically examine this design choice in Section 5.2.

SCORE uses two distinct reward channels:

$$\underbrace{R_s(r; q, D_q^e, H_q, w_q)}_{\text{solver-side report reward}} \quad \text{and} \quad \underbrace{R_e(\Gamma_q) = C(\Gamma_q)}_{\text{evaluator-side auxiliary reward}}.$$

The solver-side reward supervises report generation, while the evaluator-side reward adapts the evaluator’s rubric-selection behavior.

Solver-side report reward. For each generated report r , the evaluator produces a scalar report reward

$$R_s(r; q, D_q^e, H_q, w_q),$$

where D_q^e denotes the evidence context used by the evaluator for rubric-conditioned assessment, $H_q = \{h_1, \dots, h_m\}$ denotes the selected evaluation dimensions, and $w_q = \{w_1, \dots, w_m\}$ denotes their corresponding weights. This reward serves as the primary supervision signal for optimizing the solver.

In our implementation, the abstract report reward is instantiated as

$$R_s(r) = a(r) \cdot \left(1 + \sum_{j=1}^m w_j s_j(r; q, D_q^e) \right),$$

where $a(r)$ denotes a report-level validity or accuracy score, and $s_j(r; q, D_q^e)$ denotes the evaluator score of report r on dimension h_j under evaluator-side evidence context D_q^e . Intuitively, the solver is encouraged to generate reports that are both valid and strong along the evaluator-selected rubric dimensions.

Evaluator-side auxiliary reward. To adapt the evaluator, SCORE introduces a rubric-level auxiliary reward based on inter-report consistency. For a fixed query and evaluator-selected rubric

$$\Gamma_q := (H_q, w_q),$$

the solver generates K reports under the same query and evaluation setting. We define the raw evaluator-side auxiliary reward as

$$R_e(\Gamma_q) = C(\Gamma_q),$$

where $C(\Gamma_q)$ is the empirical consistency score computed by aggregating pairwise agreement across the generated reports. In the current implementation, $C(\Gamma_q)$ is computed as the batch mean of pairwise agreement scores over extracted report conclusions.

Evaluator updates use an EMA-centered version of this signal,

$$A_e(\Gamma_q) = R_e(\Gamma_q) - b_{t-1},$$

where b_{t-1} is the previous exponential moving average baseline of past auxiliary rewards. After computing the centered signal, the baseline is updated as

$$b_t = \beta b_{t-1} + (1 - \beta) R_e(\Gamma_q),$$

where β is the EMA decay coefficient. Thus, R_e favors rubrics under which solver behavior is more reproducible across repeated rollouts, while A_e provides the actual lower-variance signal used for REINFORCE-style evaluator updates.

Overall, SCORE forms a two-channel training loop: evaluator-derived report rewards improve the solver, while consistency-based rubric-level rewards adapt the evaluator. The solver-side reward targets report quality under the selected rubric, whereas the evaluator-side reward favors rubrics that induce more stable and reproducible solver behavior. In this sense, consistency is used as a stability-oriented auxiliary signal for rubric adaptation rather than as a direct correctness criterion.

3.6 Sequential Alternating KL-Regularized Optimization

Given the two reward channels defined in Section 3.5, training in SCORE proceeds through sequential alternating updates on the shared actor. Each step constructs a query-specific evaluation environment, generates rubric-conditioned reports, evaluates them under evaluator-side verification context, and then applies solver-side and evaluator-side updates to the shared parameters θ .

The solver-side update uses evaluator-derived report rewards R_s , which are converted into group-normalized advantages and optimized with GRPO. The evaluator-side update uses the EMA-centered auxiliary reward A_e , derived from the raw consistency reward R_e , to update the rubric-selection policy with a REINFORCE-style objective. Although these updates are computed from different trajectories and rewards, both act on the same actor and are therefore coupled through the shared parameterization.

Both updates are regularized by KL divergence against a fixed reference policy π_{ref} . In our implementation, this KL term is applied at the actor-update level rather than as an explicit reward penalty. Thus, KL regularization is not treated as a third reward channel, but as an optimization regularizer that limits excessive policy drift under shared-actor learning.

At the analysis level, we denote the unregularized role-specific objectives by $\mathcal{L}_s(\theta)$ and $\mathcal{L}_e(\theta)$, respectively. The corresponding KL-regularized objectives can be written as

$$\mathcal{L}_s^{\text{reg}}(\theta) = \mathcal{L}_s(\theta) + \lambda \text{KL}(\pi_\theta \parallel \pi_{\text{ref}}),$$

$$\mathcal{L}_e^{\text{reg}}(\theta) = \mathcal{L}_e(\theta) + \lambda \text{KL}(\pi_\theta \parallel \pi_{\text{ref}}),$$

where $\lambda > 0$ is the KL regularization coefficient.

KL regularization serves two purposes in SCORE. First, it stabilizes alternating optimization by limiting abrupt policy drift under shared-role learning. Second, it helps maintain compatibility between solver-side and evaluator-side updates on the shared actor. In this way, SCORE combines evaluator-derived report rewards, consistency-based evaluator adaptation, external grounding, shared-role learning, and KL-regularized alternating optimization into a unified training framework for long-form report generation.

4 Theoretical Analysis

We analyze two properties of SCORE: the role of consistency as an evaluator-side auxiliary signal, and the first-order optimization behavior of shared-parameter KL-regularized alternating updates.

Assumptions. We assume that: (A1) the analysis-level objectives $\mathcal{L}_s$ and $\mathcal{L}_e$ are L -smooth; (A2) the corresponding stochastic gradients are unbiased or approximately unbiased with bounded variance; and (A3) the stochastic gradients are almost surely bounded.

Consistency as rubric-induced reproducibility. Let $\Gamma_q := (H_q, w_q)$ denote the evaluator-selected rubric for query q . Suppose the solver generates K reports under the same query, evidence context, and rubric, and let $C(\Gamma_q)$ denote the resulting empirical consistency score obtained by aggregating pairwise agreement scores across these reports. In the current implementation, this is the mean pairwise consistency across generated reports. We define the raw evaluator-side auxiliary signal as

$$R_e(\Gamma_q) = C(\Gamma_q),$$

and use the centered signal

$$A_e(\Gamma_q) = R_e(\Gamma_q) - b_t$$

for evaluator updates, where b_t is an exponential moving average baseline.

Proposition 1 (Consistency estimates rubric-induced reproducibility). *Let $\mu_C(\Gamma_q)$ denote the population mean of the consistency metric under rubric Γ_q . Then*

$$\mathbb{E}[R_e(\Gamma_q)] = \mathbb{E}[\mu_C(\Gamma_q)].$$

Thus, the raw evaluator-side auxiliary signal estimates the expected reproducibility induced by rubric Γ_q under repeated solver rollouts.

Higher reproducibility suggests that the rubric induces a more stable and repeatable evaluation context for the solver, which makes it a useful auxiliary signal for evaluator-side adaptation. Centering by the EMA baseline reduces gradient variance without changing the underlying preference encoded by the raw signal. In SCORE, consistency is therefore used as a rubric-level auxiliary signal for evaluator-side adaptation.

First-order characterization of alternating updates. Let the shared-actor updates be

$$\theta' = \theta_t - \eta_s \hat{g}_s(\theta_t), \quad \theta_{t+1} = \theta' - \eta_e \hat{g}_e(\theta'),$$

where $\hat{g}_s$ and $\hat{g}_e$ are the stochastic gradients of the solver-side and evaluator-side objectives, respectively. Define the analysis surrogate

$$\tilde{\mathcal{L}}(\theta) = \mathcal{L}_s(\theta) + \alpha \mathcal{L}_e(\theta) + \lambda \text{KL}(\pi_\theta \parallel \pi_{\text{ref}}), \quad \alpha = \eta_e / \eta_s,$$

where π_{ref} is the fixed reference policy and $\lambda > 0$ is the KL regularization coefficient.

Proposition 2 (First-order characterization of alternating updates). *Under Assumptions (A1)–(A3), the expected one-step SCORE update satisfies*

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s \nabla \tilde{\mathcal{L}}(\theta_t) + \rho_t, \quad \|\rho_t\| = O(\eta_s^2).$$

This result shows that the shared-parameter solver-evaluator updates follow, to first order, a KL-regularized surrogate descent direction, with only a higher-order residual induced by sequential alternating updates. Detailed derivations are deferred to Appendix F.

5 Experiments

5.1 Experimental Setup

Training Details We train all models using the VeRL framework [36]. The evaluator and solver are updated asynchronously that the solver is updated at every training step using GRPO [35], while the evaluator is updated once every 1 steps using REINFORCE [44]. We set the rollout number to 1 for the evaluator and 8 for the solver. The maximum number of tool calls is 10, the batch size is 64, the maximum response length is 4,096 tokens, and the maximum input length is 8,192 tokens. For training, we use a local search engine backed by local WikiData, while web search is used during evaluation. GPT-5.2 [3] is applied as Meta-Harness and is invoked every 5 training steps to modify the environment according to the solver’s current performance. For the coefficient of EMA baseline, we set $\beta=0.7$. We construct the training set by collecting and preprocessing user queries from Reddit.

Table 1: Performance of open-deep-research and gpt-researcher integrated with LLaMA-3.1 and Qwen2.5 on DeepResearchBench. **Comp.**: comprehensiveness, **IF**: instruction following, **Read.**: Readability, **VC**: valid ratio of citation.

Models and Agents	Writing Quality					VC.
	Overall	Comp.	Insight	IF.	Read.	
<i>Agent: Open-deep-research (ReAct Paradigm)</i>						
Llama-3.1-8B-Inst.	27.00	24.05	22.46	30.28	36.29	22.58
+ GRPO	23.41 (-3.59)	19.89 (-4.16)	18.73 (-3.73)	26.77 (-3.51)	33.90 (-2.39)	28.57 (+5.99)
+ DPO	26.13 (-0.87)	23.11 (-0.94)	21.28 (-1.18)	29.84 (-0.44)	35.77 (-0.52)	28.00 (+5.42)
+ SCORE (Ours)	30.07 (+3.07)	30.31 (+6.26)	26.71 (+4.25)	32.25 (+1.97)	33.66 (-2.63)	30.55 (+7.97)
Qwen2.5-7B-Inst.	31.92	28.93	27.81	35.37	40.29	11.76
+ GRPO	32.67 (+0.75)	29.94 (+1.01)	29.08 (+1.27)	35.30 (-0.07)	40.87 (+0.58)	16.13 (+4.37)
+ DPO	33.91 (+1.99)	34.26 (+5.33)	30.89 (+3.08)	35.90 (+0.53)	36.85 (-3.44)	13.76 (+2.00)
+ SCORE (Ours)	34.43 (+2.51)	34.86 (+5.93)	31.38 (+3.57)	36.37 (+1.00)	37.41 (-2.88)	19.35 (+7.59)
<i>Agent: gpt-researcher (Plan-and-Execute Paradigm)</i>						
Llama-3.1-8B-Inst.	28.92	25.62	24.80	31.85	38.36	38.23
+ GRPO	16.97 (-11.95)	16.79 (-38.23)	13.12 (-11.68)	18.09 (-13.76)	23.78 (-14.58)	0.00 (-8.16)
+ DPO	33.58 (+4.66)	33.83 (+8.21)	30.64 (+5.84)	35.15 (+3.30)	36.87 (-1.49)	34.48 (-3.75)
+ SCORE (Ours)	33.89 (+4.97)	34.14 (+8.52)	30.90 (+6.10)	35.64 (+3.79)	37.29 (-1.07)	52.38 (+14.15)
Qwen2.5-7B-Inst.	32.00	29.20	28.70	34.67	39.61	36.11
+ GRPO	30.03 (-1.97)	27.52 (-1.68)	27.39 (-1.31)	31.07 (-3.60)	38.37 (-1.24)	51.61 (+15.50)
+ DPO	35.85 (+3.85)	36.22 (+7.02)	33.64 (+4.94)	37.15 (+2.48)	38.07 (-1.54)	30.80 (-5.31)
+ SCORE (Ours)	35.91 (+3.91)	36.15 (+6.95)	33.12 (+4.42)	37.60 (+2.93)	38.93 (-0.68)	66.15 (+30.04)

Baselines and Models We implement SCORE on top of two capable base LLMs, Qwen2.5-7B-Instruct [32] and Llama-3.1-8B-Instruct [16]. For comparison, we train these models using standard GRPO and DPO to serve as baselines. The post-trained models are subsequently evaluated within two representative search agent systems to assess their downstream capabilities. Specifically, we employ open-deep-research [2] as a baseline for the ReAct paradigm and gpt-researcher [1] as a baseline for the plan-and-execute paradigm. To maintain experimental consistency, a single, unified model is deployed across all functional modules within these respective agents.

Benchmarks and Evaluation Metrics To rigorously evaluate our approach, we employ two benchmarks designed for deep research tasks which are DeepResearchBench [13] and DeepResearchEval [43]. DeepResearchBench assesses agent performance across multi-dimensions including comprehensiveness, insight, instruction following, readability, and citation validity. In contrast, DeepResearchEval utilizes a dynamic evaluation metric, which aligns closely with our methodological perspective. Both frameworks ultimately adopt GPT-5.2 [57] for the final evaluation.

5.2 Experimental Results

Main Results As illustrated in Table 1, SCORE demonstrates a substantial and robust improvement in analytical depth and valid citation across all agents. Basic reinforcement learning (RL) methods such as GRPO and DPO suffer from a catastrophic degradation in at least one dimension, occasionally collapsing to exactly zero in Plan-and-Execute agents when integrating LLaMA-3.1-8B-Instruct. This phenomenon is driven by Lazy Likelihood Displacement (LLD), a condition where the likelihood of correct reasoning trajectories decays during tool-integrated RL, ultimately causing training collapse [10]. Furthermore, agent architectural magnify this instability, as decoupled long-horizon frameworks like Plan-and-Execute exhibit premature collapse and severe reward hacking compared to step-by-step ReAct policies [9]. In contrast, SCORE maintains stable performance by utilizing a co-evolving evaluator that provides dense, consistent constraints through dynamic evaluation rubrics. Notably, we observe a consistent decline in readability across nearly all training methods. We hypothesize this a fundamental trade-off because models are optimized to synthesize comprehensive analysis and rigorously cite evidence, the generated report inherently incorporates denser academic vocabulary

and more complex logical structures, thereby sacrificing conversational simplicity for analytical rigor which might be similar to the trade-off between diversity and factuality [25].

Module Ablation Study Refer to Figure 3, we observe that removing any single module severely degrades the LLMs’ comprehensiveness, demonstrating that generating well-rounded research reports requires the synergistic operation of the entire framework. Freezing the solver leads to a massive decline in citation validity. The internal consistency reward for evaluator brings reward hacking, where the evaluator learns to construct superficial rubrics that elicit highly consistent text from the static solver while actively avoiding high-variance behaviors like rigorous external citation. Furthermore, our complete method exhibits a slightly lower readability score compared to the ablated baselines, which we attribute to the inherent trade-off between natural language coherence and academic formatting. Base models tend to produce fluent but unstructured text, whereas our co-evolutionary framework forces the model to generate deeper reports with complex citation syntax, thus sacrificing superficial readability for factual rigor.

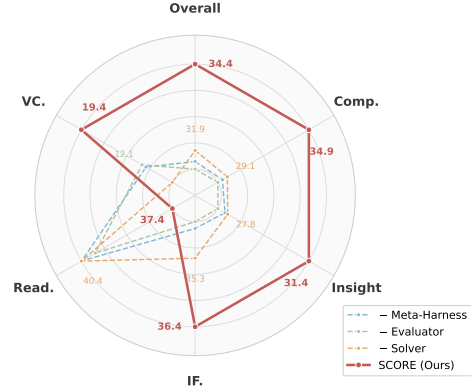


Figure 3: Ablation study for individual modules on DeepResearchBench. **Comp.:** comprehensiveness, **IF:** instruction following, **Read.:** Readability, **VC:** valid ratio of citation.

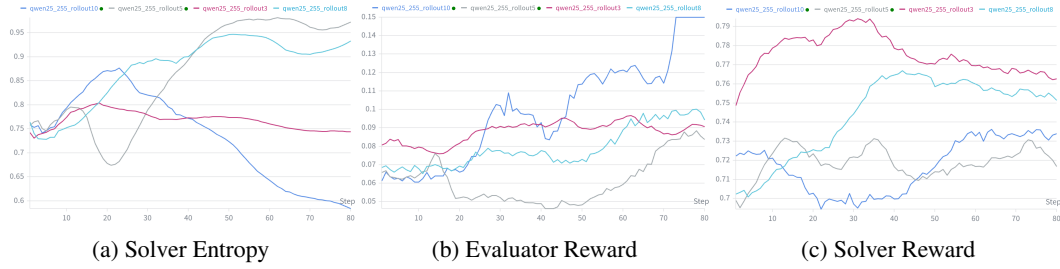


Figure 4: Training Details

Experiments on Rollout Number We investigate the impact of the rollout number K in the training progress. Figure 4 demonstrates that extreme K affect training procedure that small $K = 3$ fails to yield meaningful reward improvements, while large $K = 10$ triggers a severe model collapse reflected by sudden drop of entropy [37]. Small K fails to provide the solver with statistically robust reward due to insufficient ranking samples while for the evaluator, fewer rollouts yield unreliable pairwise consistency estimates, resulting in noisy REINFORCE gradients that prevent stable dimension policy updates. Conversely, an overly large K yields highly precise advantage estimates for both the solver and the evaluator, accelerating their updates. Under the self-evolving setting, this mutual acceleration compounds policy drift and may trigger rapid model collapse.

Case Study As illustrated in Figure 6 in Appendix G.2, different types of research queries inherently require distinct evaluation criteria. For instance, legal queries demand absolute precision and strict citation of statutory texts, whereas open-ended questions require the aggregation of diverse perspectives to maintain fairness and objectivity. A generic, static rubric fails to capture these critical nuances, often assigning acceptable scores to reports that are superficially fluent but fundamentally flawed. In contrast, our adaptive evaluation framework dynamically introduces specialized evaluation dimensions tailored to each query. This mechanism ensures that the generated reports are accurately penalized or rewarded according to the specific rigor demanded by the task.

6 Conclusion

Generating comprehensive reports for deep research is a highly complex task, and the inherent difficulty of evaluating these open-ended reports severely hinders the training process. We observe a

strong positive correlation between a model’s evaluation and generation capabilities. Based on this insight, we propose SCORE, a self-evolving training framework that jointly optimizes an evaluator and a solver. Because the entire training procedure fundamentally relies on unverifiable rewards, we introduce an external meta-harness to impose necessary environmental constraints on the evaluator. This module simultaneously guides the evaluator to actively explore a broader range of relevant evaluation dimensions throughout the training phase. Finally, experimental results demonstrate that our approach achieves substantial performance improvements while requiring only a remarkably small amount of training data.

References

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A Limitations

Despite the effectiveness of our self-evolving framework, we acknowledge several limitations in our current study. First, the unique nature of deep research report generation demands exceptional instruction-following capabilities from the base language models. Models with weaker foundational abilities may struggle with dynamic rubrics generation and tool call. Second, our training paradigm relies heavily on entirely unverifiable rewards. This lack of ground-truth makes the model susceptible to policy collapse during the optimization process. Consequently, careful hyperparameter tuning and strict early stopping mechanisms are necessary to ensure training stability. Even though our framework successfully provides differentiated reward signals for specific criteria, aggressively optimizing for one dimension can sometimes lead to performance degradation in another. Balancing these competing objectives simultaneously remains a challenging problem for future exploration.

B Future Work

According to our experiments, we observe that inherent trade-offs still exist during the training process for deep research report generation. We hypothesize that this issue is primarily caused by sparse reward signals resulting from exceptionally long agent trajectories. To address this challenge, future work will explore phased training paradigms. By decomposing the deep research generation process into distinct functional stages and assigning intermediate rewards, we aim to resolve the long-standing credit assignment problem that plagues end-to-end policy optimization.

C LLM Usage Declaration

LLMs are employed to improve the grammar and language quality of the paper. LLMs are also applied to check the correctness and clarity of the proofs. In the experiments, open-source LLMs were used for post-train models, while closed-source LLMs were used as judges for assessment. In addition, a coding agent was used to assist in developing parts of the training framework.

D Licenses

We use the following models in the experiments:

- Qwen2.5-7B-Instruct, Qwen2.5-72B-Instruct [32] and Qwen3-8B [47] with License *Apache 2.0*.
- Llama-3.1-8B-Instruct [16] with License *Llama 3.1 Community*.
- GLM-4.7-Flash [42] and DeepSeek-V3.2 [8] with License *mit*.
- Gemini-2.5-pro [7] with License *Apache 2.0* and GPT-5.2 [3] with proprietary license.

E Algorithm

Algorithm 1 summarizes the main shared-actor evaluator–solver training loop, while Algorithm 2 details the evaluator-side consistency reward construction and rubric adaptation step. In the current implementation, PAIRWISECONSISTENCY is instantiated as the mean pairwise TF-IDF cosine similarity over extracted rollout conclusions.

Algorithm 1 SCORE: Evaluator–Solver Self-Play with Consistency-Driven Dimension Learning

Require: Shared actor parameters θ ; evaluator role policy $\pi_e(\cdot; \theta)$; solver role policy $\pi_s(\cdot; \theta)$; dimension pool $\mathcal{D}$; solver rollout size K ; batch size B ; evaluator update period U ; harness refresh period M ; minimum verification/search depth τ_s ; KL coefficient λ_{KL}

Ensure: Trained shared actor parameters θ

- 1: Initialize DimensionScheduler $\mathcal{S}$ over $\mathcal{D}$
- 2: Initialize Meta-Harness state $\mathcal{M}_0 \leftarrow \mathcal{M}_{\text{default}}$
- 3: Initialize evaluator-side EMA baseline $b_0 \leftarrow 0$
- 4: **for** each training step $t = 1, 2, \dots, T_{\text{max}}$ **do**
 - $\triangleright$ Phase 1: Refresh scheduler and harness constraints
 - 5: $(\mathcal{R}_t, \mathcal{A}_t) \leftarrow \mathcal{S}.\text{GETCONSTRAINTS}()$ $\triangleright$ retrieve the current required / avoided dimensions and exploration hints
 - 6: Obtain current dimension and process constraints from $\mathcal{M}_{t-1}$ $\triangleright$ the meta-harness defines the admissible rubric space for this step
 - $\triangleright$ Phase 2: Evaluator samples a rubric
 - 7: Sample query batch $\{q_i\}_{i=1}^B$
 - 8: Build evaluator rubric-generation prompt x_t^{rubric} from $(\mathcal{R}_t, \mathcal{A}_t, \mathcal{M}_{t-1})$
 - 9: Sample rubric
$$(H_t, w_t), \log \pi_e(H_t, w_t \mid x_t^{\text{rubric}}; \theta) \sim \pi_e(\cdot \mid x_t^{\text{rubric}}; \theta)$$
$$\triangleright \text{the evaluator chooses what dimensions to emphasize and how to weight them}$$
$$\triangleright \text{Phase 3: Solver performs rubric-conditioned rollouts}$$
 - 10: **for** each query q_i , each rollout $k = 1, \dots, K$ **do**
 - 11: Generate report trajectory and final report
$$y_i^k \sim \pi_s(\cdot \mid q_i, H_t, w_t; \theta)$$
 - 12: Record solver-side evidence / tool observations $D_{q_i, k}^s$ $\triangleright$ the solver searches, interacts with tools, and produces a long-form report
 - 13: **end for**
$$\triangleright \text{Phase 4: Evaluator assesses reports}$$
 - 14: Optionally build evaluator-side evidence contexts $\{D_{q_i}^e\}_{i=1}^B$ through constrained retrieval / verification $\triangleright$ the evaluator may gather its own evidence for rubric-conditioned assessment
 - 15: **for** each query q_i , each rollout $k = 1, \dots, K$ **do**
 - 16: Compute evaluator-side report reward
$$s_i^k \leftarrow \text{EVALSCORE}(q_i, y_i^k, H_t, w_t, D_{q_i}^e, \mathcal{M}_{t-1}; \theta)$$
 - 17: **if** evaluator verification/search depth for (i, k) is below τ_s **then** set $s_i^k \leftarrow 0$ $\triangleright$ discard under-verified rewards to avoid noisy solver supervision
 - 18: **end for**
$$\triangleright \text{Phase 5: Solver-side GRPO update}$$
 - 19: **for** each query q_i **do**
 - 20: $\mu_i \leftarrow \text{mean}_k s_i^k, \quad \sigma_i \leftarrow \text{std}_k s_i^k$
 - 21: $\hat{A}_i^k \leftarrow (s_i^k - \mu_i) / (\sigma_i + \epsilon)$ $\triangleright$ group-normalized rewards define solver-side advantages
 - 22: **end for**
 - 23: Update shared actor parameters θ using the solver-side GRPO objective with KL regularization against π_{ref}
$$\triangleright \text{Phase 6: Evaluator-side consistency update}$$
 - 24: Compute the evaluator-side consistency signal and update the evaluator role as detailed in Algorithm 2 $\triangleright$ the evaluator is rewarded when its rubric induces more reproducible solver behavior
 - $\triangleright$ Phase 7: Refresh scheduler and meta-harness
 - 25: $\mathcal{S}.\text{UPDATESTATS}(H_t, \{C_i\}_{i=1}^B)$ $\triangleright$ record which dimensions produced more reproducible rollouts
 - 26: **if** $t \bmod M = 0$ **then**
 - 27: Collect harness statistics $\mathcal{T}_t$
 - 28: $\mathcal{M}_t \leftarrow \text{HARNESS.GENERATE}(\mathcal{T}_t, \mathcal{M}_{t-1}, \mathcal{M}_{\text{default}})$ $\triangleright$ periodically refresh the admissible evaluation space
 - 29: **else**
 - 30: $\mathcal{M}_t \leftarrow \mathcal{M}_{t-1}$
 - 31: **end if**
 - 32: **end for**

Algorithm 2 Evaluator-Side Consistency Reward and Rubric Update

Require: Rollout reports $\{y_i^k\}_{i=1,\dots,B}^{k=1,\dots,K}$; sampled rubric (H_t, w_t) ; rubric prompt x_t^{rubric} ; previous EMA baseline b_{t-1} ; baseline decay β ; evaluator scale α ; evaluator update period U

Ensure: Updated EMA baseline b_t , consistency statistics $\{C_i\}_{i=1}^B$, and optionally updated shared actor parameters θ

▷ *Phase 1: Build a batch-level consistency signal*

- 1: **for** each query q_i **do**
- 2: Extract rollout conclusions $\{c_i^k\}_{k=1}^K$ from $\{y_i^k\}_{k=1}^K$ ▷ measure agreement at the conclusion level rather than at the token level
- 3: Compute empirical consistency

$$C_i \leftarrow \text{PAIRWISECONSISTENCY}(\{c_i^k\}_{k=1}^K)$$

4: **end for**

5: Compute evaluator-side raw signal

$$R_{\text{eval}} \leftarrow \frac{1}{B} \sum_{i=1}^B C_i$$

▷ the raw signal favors rubrics under which solver rollouts are more reproducible

▷ *Phase 2: Center the signal with an EMA baseline*

6: Compute centered evaluator-side signal

$$\hat{A}_{\text{eval}} \leftarrow R_{\text{eval}} - b_{t-1}$$

7: Update EMA baseline

$$b_t \leftarrow \beta b_{t-1} + (1 - \beta) R_{\text{eval}}$$

▷ the EMA baseline stabilizes REINFORCE updates without changing the raw preference

▷ *Phase 3: Update the evaluator role*

8: **if** $t \bmod U = 0$ **and** evaluator updates are enabled **then**

9: Update shared actor parameters θ using the evaluator-side REINFORCE objective

$$\theta \leftarrow \theta + \eta \alpha \hat{A}_{\text{eval}} \nabla_{\theta} \log \pi_e(H_t, w_t \mid x_t^{\text{rubric}}; \theta)$$

▷ rubrics that induce more reproducible solver behavior become more likely

10: **end if**

F Detailed Proofs

This appendix provides derivations for the results stated in Section 4. Throughout, let $\theta \in \mathbb{R}^d$ denote the shared actor parameters, and let

$$\tilde{\mathcal{L}}(\theta) = \mathcal{L}_s(\theta) + \alpha \mathcal{L}_e(\theta) + \lambda \text{KL}(\pi_{\theta} \parallel \pi_{\text{ref}})$$

denote the KL-regularized surrogate potential.

F.1 Derivation of Proposition 1

Recall that $\Gamma_q := (H_q, w_q)$ denotes the evaluator-selected rubric for query q . Suppose the solver generates K reports under the same query, evidence context, and rubric. Let

$$c_{ij}(\Gamma_q), \quad 1 \leq i < j \leq K,$$

denote the pairwise agreement score between reports r_i and r_j under rubric Γ_q , and let

$$M = \binom{K}{2}$$

be the number of report pairs. In the current implementation, the empirical consistency score is defined as

$$C(\Gamma_q) = \frac{1}{M} \sum_{1 \leq i < j \leq K} c_{ij}(\Gamma_q).$$

The raw evaluator-side auxiliary signal is

$$R_e(\Gamma_q) = C(\Gamma_q).$$

Let

$$\mu_C(\Gamma_q) := \mathbb{E}[c_{ij}(\Gamma_q)]$$

denote the population mean of the consistency metric under rubric Γ_q . Then

$$\mathbb{E}[R_e(\Gamma_q)] = \mathbb{E}[C(\Gamma_q)] = \mathbb{E}\left[\frac{1}{M} \sum_{1 \leq i < j \leq K} c_{ij}(\Gamma_q)\right].$$

By linearity of expectation,

$$\mathbb{E}[R_e(\Gamma_q)] = \frac{1}{M} \sum_{1 \leq i < j \leq K} \mathbb{E}[c_{ij}(\Gamma_q)] = \frac{1}{M} \sum_{1 \leq i < j \leq K} \mu_C(\Gamma_q) = \mu_C(\Gamma_q).$$

Taking expectation over the induced rubric distribution yields

$$\mathbb{E}[R_e(\Gamma_q)] = \mathbb{E}[\mu_C(\Gamma_q)].$$

This proves Proposition 1.

In implementation, evaluator updates use the centered signal

$$A_e(\Gamma_q) = R_e(\Gamma_q) - b_t,$$

where b_t is an exponential moving average baseline. Since b_t is independent of the current sampled rubric action, subtracting it reduces gradient variance without changing the underlying expected preference encoded by $R_e(\Gamma_q)$.

F.2 Derivation of Proposition 2

The shared-actor updates are

$$\theta' = \theta_t - \eta_s \hat{g}_s(\theta_t), \quad \theta_{t+1} = \theta' - \eta_e \hat{g}_e(\theta').$$

Hence,

$$\theta_{t+1} - \theta_t = -\eta_s \hat{g}_s(\theta_t) - \eta_e \hat{g}_e(\theta').$$

Taking conditional expectation with respect to θ_t gives

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s \mathbb{E}[\hat{g}_s(\theta_t) \mid \theta_t] - \eta_e \mathbb{E}[\hat{g}_e(\theta') \mid \theta_t].$$

Under Assumption (A2),

$$\mathbb{E}[\hat{g}_s(\theta_t) \mid \theta_t] \approx \nabla \mathcal{L}_s(\theta_t), \quad \mathbb{E}[\hat{g}_e(\theta') \mid \theta_t] \approx \nabla \mathcal{L}_e(\theta').$$

Thus,

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] \approx -\eta_s \nabla \mathcal{L}_s(\theta_t) - \eta_e \nabla \mathcal{L}_e(\theta').$$

By Assumption (A1), $\mathcal{L}_e$ is L -smooth, so

$$\|\nabla \mathcal{L}_e(\theta') - \nabla \mathcal{L}_e(\theta_t)\| \leq L \|\theta' - \theta_t\|.$$

From the first update,

$$\theta' - \theta_t = -\eta_s \hat{g}_s(\theta_t),$$

and by Assumption (A3), $\|\hat{g}_s(\theta_t)\| \leq G$. Therefore,

$$\|\theta' - \theta_t\| \leq \eta_s G,$$

which implies

$$\|\nabla \mathcal{L}_e(\theta') - \nabla \mathcal{L}_e(\theta_t)\| \leq L \eta_s G.$$

Hence we may write

$$\nabla \mathcal{L}_e(\theta') = \nabla \mathcal{L}_e(\theta_t) + \delta_t, \quad \|\delta_t\| = O(\eta_s).$$

Substituting this into the expected update yields

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s \nabla \mathcal{L}_s(\theta_t) - \eta_e \nabla \mathcal{L}_e(\theta_t) - \eta_e \delta_t.$$

Using $\alpha = \eta_e / \eta_s$, we obtain

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s \nabla (\mathcal{L}_s(\theta_t) + \alpha \mathcal{L}_e(\theta_t)) + \rho_t^{(0)},$$

where

$$\rho_t^{(0)} = -\eta_e \delta_t, \quad \|\rho_t^{(0)}\| = O(\eta_s^2).$$

Including KL regularization against the fixed reference policy π_{ref} , we define

$$\tilde{\mathcal{L}}(\theta) = \mathcal{L}_s(\theta) + \alpha \mathcal{L}_e(\theta) + \lambda \text{KL}(\pi_\theta \parallel \pi_{\text{ref}}).$$

The same argument then gives

$$\mathbb{E}[\theta_{t+1} - \theta_t \mid \theta_t] = -\eta_s \nabla \tilde{\mathcal{L}}(\theta_t) + \rho_t, \quad \|\rho_t\| = O(\eta_s^2),$$

which proves Proposition 2.

F.3 Summary

The above derivations establish two facts used in the main text:

1. the raw evaluator-side consistency signal estimates the expected reproducibility induced by a rubric under repeated solver rollouts;
2. the shared-parameter solver-evaluator alternating updates follow, to first order, a KL-regularized surrogate descent direction, with only a higher-order residual due to sequential updating.

G Experiments

G.1 Training Details

Table 2: Main training hyperparameters in SCORE.

Category	Configuration
Learning rate	1×10^{-6}
Warmup	5 steps, warmup ratio 0.05
Training Data Size	255 for Qwen2.5-7B and 2,380 for Llama-3.1-8B
KL regularization	Coefficient 0.05, low-variance KL, applied in actor loss only
Batch size	Training batch size 64; PPO mini-batch size 32; PPO micro-batch size 4 per GPU
Rollouts	Solver rollout number 8; evaluator rollout number 1
Context and generation limits	Maximum input length 8192; maximum response length 4096; evaluator maximum generation length 2048
Interaction budget	Maximum assistant turns 10
Evaluator update	EMA baseline decay 0.7; update frequency every step; temperature 0.8
Meta-harness	Refresh frequency every 5 training steps
Training retrieval	Local static retrieval backend use wikidata
Compute	Single-node training with 4 H800, 100 CPU cores, and 1 TB host memory
Systems optimization	FSDP, Ray orchestration, gradient checkpointing, parameter offloading, optimizer offloading
Training schedule	50-120 steps

G.2 Experimental Results

Results of DeepResearchEval We also evaluate the search agents on DeepResearchEval [43], where evaluation rubrics are customized for each specific query. The results shown in Table 3 strongly corroborate our previous observations that fundamental GRPO methods suffer severe performance

Table 3: Performance of open-deep-research and gpt-researcher integrated with LLaMA-3.1 and Qwen2.5 on DeepResearchEval. **Cove**: Breadth, depth, and relevance of coverage. **Insight**: Depth, originality, logic, and value of analysis. **IF**: Accuracy in meeting all requirements. **Clar**: Clarity, fluency, structure, and ease of understanding. **Meta**: Query-specific meta-evaluation dimensions generated dynamically for each task.

Models and Agents	Cove	Insight	IF	Clar	Meta
<i>Agent: Open-deep-research (ReAct Paradigm)</i>					
Llama-3.1-8B-Inst.	3.7	3.2	4.5	5.1	2.6
+ GRPO	1.2 (-2.5)	1.3 (-1.9)	1.4 (-3.1)	1.8 (-3.3)	1.0 (-1.6)
+ DPO	4.3 (+0.6)	4.3 (+1.1)	4.7 (+0.2)	5.2 (+0.1)	3.5 (+0.9)
+ SCORE (Ours)	4.3 (+0.6)	4.4 (+1.2)	4.7 (+0.2)	5.2 (+0.1)	3.5 (+0.9)
Qwen2.5-7B-Inst.	4.4	3.9	5.2	5.4	3.2
+ GRPO	4.4 (+0.0)	4.0 (+0.1)	5.2 (+0.0)	5.4 (+0.0)	3.3 (+0.1)
+ DPO	4.6 (+0.2)	4.1 (+0.2)	5.6 (+0.4)	5.8 (+0.4)	3.4 (+0.2)
+ SCORE (Ours)	6.3 (+1.9)	5.9 (+2.0)	7.0 (+1.8)	6.6 (+1.2)	5.0 (+1.8)
<i>Agent: gpt-researcher (Plan-and-Execute Paradigm)</i>					
Llama-3.1-8B-Inst.	3.9	3.5	4.8	5.6	2.8
+ GRPO	2.5 (-1.4)	2.8 (-0.7)	2.7 (-2.1)	3.7 (-1.9)	2.0 (-0.8)
+ DPO	2.4 (-1.5)	2.8 (-0.7)	2.6 (-2.2)	3.5 (-2.1)	2.0 (-0.8)
+ SCORE (Ours)	5.9 (+2.0)	5.6 (+2.1)	6.9 (+2.1)	6.9 (+1.3)	4.6 (+1.8)
Qwen2.5-7B-Inst.	4.5	4.1	5.7	5.9	3.4
+ GRPO	4.6 (+0.1)	4.1 (+0.0)	5.6 (-0.1)	5.8 (-0.1)	3.4 (+0.0)
+ DPO	6.1 (+1.6)	5.8 (+1.7)	6.8 (+1.1)	6.5 (+0.6)	4.8 (+1.4)
+ SCORE (Ours)	6.5 (+2.0)	6.0 (+1.9)	7.5 (+1.8)	7.0 (+1.1)	5.1 (+1.7)

degradation, whereas our framework consistently achieves the highest scores across all dimensions, including the highly challenging query-specific Meta metrics. Interestingly, while generic readability declines on DeepResearchBench benchmarks, our method achieves consistent gains in the ‘‘Clarity’’ metric on DeepResearchEval. This divergence supports our trade-off hypothesis, confirming that the generated texts exchange conversational simplicity for a highly structured and analytically clear presentation, which is heavily rewarded when evaluating complex, query-specific research explanations.

Training Data Size We apply different training data size in SCORE. While the co-evolving evaluator successfully stabilizes training on smaller datasets, increasing the data scale accelerates model collapse. The reason is both of the solver and evaluator rely on instinct reward rather than strictly verifiable external ground truth. With limited training data, this exact internal feedback effectively sharpens the model’s correct priors and provides necessary dense constraints. However, as the dataset size increases significantly, the solver progressively discovers blind spots in the evaluator and over-optimizes against these proxy rewards. This dynamic inevitably forces the optimization process to saturate, degrading the factual accuracy as the solver exploits the distributional limits of the proxy feedback. As illustrated by Figure 5, the solver entropy collapses prematurely at large scales, followed by a classic rise-then-fall trajectory in the training rewards. This observation aligns perfectly with recent theoretical findings on unsupervised reinforcement learning, confirming that intrinsic reward mechanisms eventually amplify self-reinforcing hallucinations when exposed to extensive data [17].

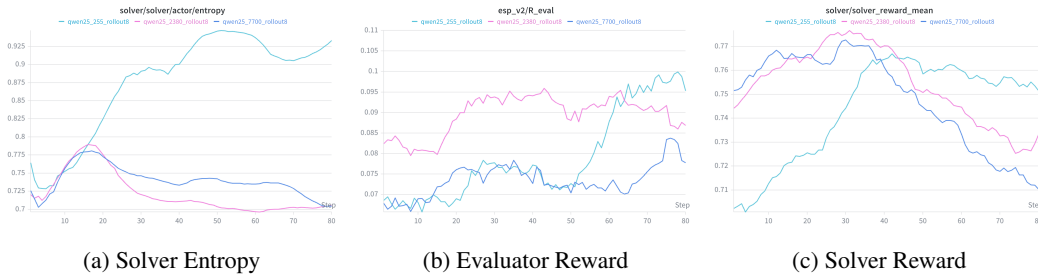


Figure 5: Training Details

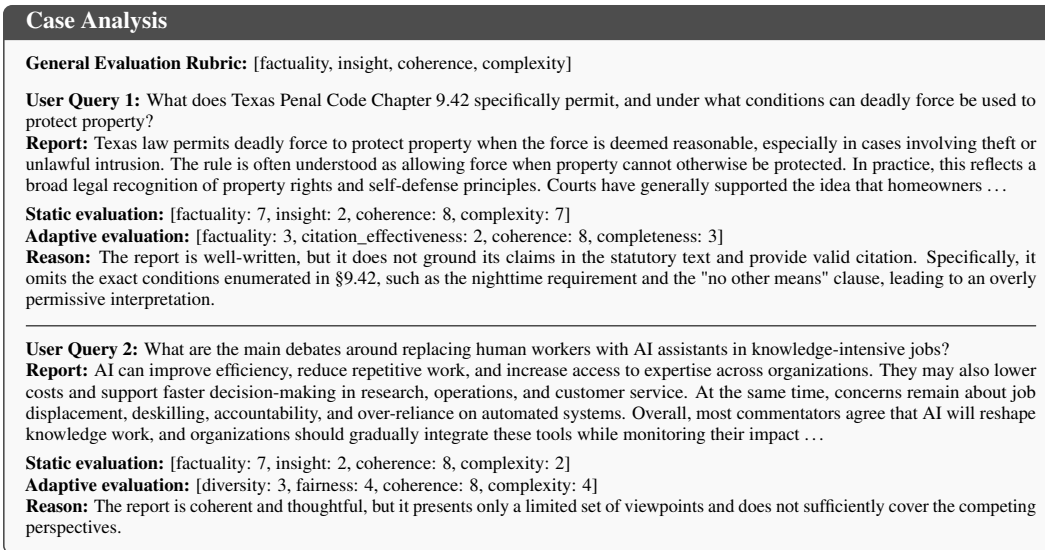


Figure 6: Adaptive evaluation vs static evaluation of different queries.

Case Study Figure 6 demonstrates the stark contrast in scoring behaviors between static and adaptive evaluations. Under the static rubric, the solver successfully exploits the generic "coherence" metric, securing a near-perfect score of 8 in both scenarios by generating superficially fluent prose. However, this high structural score masks critical omissions in substantive content. When our adaptive framework is applied, it immediately synthesizes query-specific constraints to interrogate the text. In the legal scenario, the dynamic evaluation criteria focus on whether the report grounds its claims in specific statutory prerequisites. Similarly, in the discussion scenario, the metrics expose the report's failure to synthesize competing viewpoints, despite its seemingly thoughtful tone. This dynamic assessment effectively prevents the model from prioritizing elegant phrasing to secure high scores while neglecting factual accuracy or other critical dimensional requirements.

H Prompts

```
You will be provided with a reference and some statements. Please determine
whether each statement is 'supported', 'unsupported', or 'unknown' with
respect to the reference. Please note:
First, assess whether the reference contains any valid content. If the
reference contains no valid information, such as a 'page not found'
message, then all statements should be considered 'unknown'.
If the reference is valid, for a given statement: if the facts or data it
contains can be found entirely or partially within the reference, it is
considered 'supported' (data accepts rounding); if all facts and data in
the statement cannot be found in the reference, it is considered '
unsupported'.

You should return the result in a JSON list format, where each item in the
list contains the statement's index and the judgment result, for example:
[
  {{
    "idx": 1,
    "result": "supported"
  }},
  {{
    "idx": 2,
    "result": "unsupported"
  }}
]

Below are the reference and statements:
<reference>
{reference}
</reference>

<statements>
{statements}
</statements>

Begin the assessment now. Output only the JSON list, without any
conversational text or explanations.
```

Listing 1: Prompt for citation checking.

```
<system_role>
You are an experienced research article evaluation expert. Your task is to
evaluate a deep research article based on the provided '<task>' across
four dimensions: Comprehensiveness, Insight, Instruction Following, and
Readability.
</system_role>

<task>
"{task_prompt}"
</task>

<instruction>
Please deeply analyze the task and evaluate the research report based on the
following specific criteria for each dimension:

1. Comprehensiveness (Information Coverage)
- Evaluate the breadth, depth, and relevance of information coverage.
- Ensure the article covers key information areas, perspectives, and depths
necessary to achieve comprehensive market or situational analysis.
- Check if it thoroughly examines all relevant factors (e.g., technical,
economic, social, geographical) and avoids omitting critical facets of the
topic.

2. Insight (Analytical Depth)
- Evaluate the depth, originality, logic, and value of the analysis and
conclusions.
```

- Assess whether the analysis goes beyond a superficial listing of factors to uncover core drivers, subtle mechanisms, and second-order effects.
- Examine if the conclusions and recommendations are logically derived from the analysis, offer unique perspectives, and provide forward-looking or actionable strategic value.

3. Instruction Following (Responsiveness & Relevance)

- Evaluate whether the report accurately, completely, and directly responds to all specific instructions, questions, and core objectives within the ‘<task>’.
- Check for strict adherence to scope limitations (e.g., specific geography, time, or subject constraints) without significant deviation into irrelevant topics.

4. Readability (Presentation Quality)

- Evaluate the clarity of structure, fluency of language, effectiveness of data presentation, and overall ease of understanding.
- Assess structural logic (clear framework and logical flow), language precision (fluent, grammatically correct, and appropriate use of terminology), and cohesive paragraph transitions.
- Check the clarity and accuracy of data presentation, including the effective use of visualizations (charts, tables) and the highlighting of key findings to reduce reader fatigue.

</instruction>

Listing 2: Prompt for writing quality evaluation

```
[System Prompt]
You are an external researcher. Adjust the evaluator harness based on recent
training statistics. Do not choose the evaluator's final dimensions or
weights directly.

To request search from the evaluator, set require_search=true in
process_constraints, and explicitly state in notes that the evaluator
should use the search tool by outputting <search>query</search> tags.

[User Prompt]
Task: Generate the next evaluator harness H_t.
Goal: Update evaluator constraints without directly choosing concrete
dimensions or weights for the evaluator. You may adjust
dimension_constraints and process_constraints and dimension_pool.

## Statistics:
{Training Statistics JSON}

## Output Format:
{
  "dimension_constraints": {
    "k": number of dimensions,
    "must_include": [list of required dimension names],
    "weight_min": minimum weight for a dimension,
    "weight_max": maximum weight for a dimension
  },
  "process_constraints": {
    "require_gap_analysis": true/false,
    "require_verification": true/false,
    "citation_cross_check": true/false,
    "min_number_of_tool_call": n
  },
  "notes": "additional process compliance instructions"
}
```

Listing 3: Prompt for meta-harness

```
[System Prompt]
You are a professional evaluator for deep research reports.
You will receive one query and multiple candidate reports generated by the
solver.
```

```

Your job is to evaluate every report with a search-assisted review process.

You must complete 4 stages:
1. {must_include} is always required as the primary scoring dimension. Select
   exactly {k-1} additional dimensions from the remaining pool and assign
   weights, each additional dimension's weight must be within [{Weight Min},
   {Weight Max}]
2. Verify key claims and citations in each report using the available search
   tool.
3. Perform independent research from the original query to identify important
   gaps not covered by each report.
4. Score each report on every selected dimension using the evidence you found
   .

## Tool-use protocol:
- When you need external evidence, you can use search tools with search query
  in this format:
  <search>your search query here</search>
- The content inside <search> must be only the query text. Do not output JSON
  , function-call syntax, XML attributes, bullet lists, or multiple queries.
- After a search, the system will return retrieved evidence inside <
  observation>...</observation>.
- You may issue multiple search queries as needed before producing the final
  XML.

Return the final result in the following XML format ONLY AFTER you finish
your search and evaluation:
<dimensions>
  <dim name="[Base Dimension]"/>
  <dim name="[Selected Dimension]" weight="[Weight]"/>
</dimensions>
<report_evaluations>
  <report id="[Report ID]">
    <verification>your verification process here</verification>
    <gap_analysis>your gap analysis here</gap_analysis>
    <scores>
      <score dim="[Base Dimension]" value="[Score]" reason="..." />
      <score dim="[Selected Dimension]" value="[Score]" reason="..." />
    </scores>
  </report>
</report_evaluations>

```

Listing 4: Prompt for evaluator

```

[System Prompt]
You are a professional deep researcher. Solve the user's request by
iteratively searching public information and synthesizing evidence.

## Rules:
1. Think inside <think> and </think> before each action and after each new
  search result.
2. When external evidence is needed, emit exactly one standalone search query
  . Use the exact format <search>your query here</search> to issue a search.
3. The content inside <search> must be only the query text. Do not output
  JSON, tool names, XML attributes, bullet lists, or multiple queries in one
  search block.
4. Prefer natural web search queries: start broad, then narrow to
  verification, recent developments, entity-specific facts, or missing
  perspectives.
5. Prefer multiple sequential searches over one overloaded query.
6. Use search to verify important claims, gather evidence, and close
  information gaps. Stop searching once you can answer confidently.
7. The environment will return search results between <observation> and </
  observation>.
8. When you have enough evidence, write the final response only inside <
  answer> and </answer>.
9. In the final answer, synthesize the evidence, note uncertainty when
  necessary, and include source mentions or a short Sources section.

```

Listing 5: Prompt for solver